

Typing Rules, Domain Semantics, and Pi Injectivity for Dependent Type Theory with $\mathbf{U} : \mathbf{U}$ and $\mathbf{Prop} : \mathbf{U}$

1 Judgments

We work with three mutually defined judgment forms:

$\Gamma \text{ ctx}$	Γ is a well-formed context
$\Gamma \vdash M : A$	M has type A in context Γ
$\Gamma \vdash M = N : A$	M and N are definitionally equal at type A

Contexts are snoc-lists: $\Gamma ::= \cdot \mid \Gamma, x:A$. We write $M\sigma$ for the result of applying a substitution σ to a term M , and $[t]$ for the substitution $\Gamma \rightarrow \Gamma, x:A$ extending the identity by t , so that $B[a]$ means $B(\text{id}, a)$.

2 Well-formed contexts

$$\frac{}{\cdot \text{ ctx}} \qquad \frac{\Gamma \vdash A : \mathbf{U}}{\Gamma, x:A \text{ ctx}}$$

3 Typing rules

$$\begin{array}{c} \text{VAR} \\ \frac{\Gamma \text{ ctx} \quad (x:A) \in \Gamma}{\Gamma \vdash x : A} \end{array} \qquad \begin{array}{c} \text{CONV} \\ \frac{\Gamma \vdash M : A \quad \Gamma \vdash A = B : \mathbf{U} \quad \Gamma \vdash B : \mathbf{U}}{\Gamma \vdash M : B} \end{array} \qquad \begin{array}{c} \mathbf{U} \\ \frac{\Gamma \text{ ctx}}{\Gamma \vdash \mathbf{U} : \mathbf{U}} \end{array}$$

$$\begin{array}{c} \text{PI} \\ \frac{\Gamma \vdash A : \mathbf{U} \quad \Gamma, x:A \vdash B : \mathbf{U}}{\Gamma \vdash \Pi(x:A)B : \mathbf{U}} \end{array} \qquad \begin{array}{c} \text{LAM} \\ \frac{\Gamma \vdash A : \mathbf{U} \quad \Gamma, x:A \vdash B : \mathbf{U} \quad \Gamma, x:A \vdash M : B}{\Gamma \vdash \lambda(x:A)M : \Pi(x:A)B} \end{array}$$

$$\begin{array}{c} \text{APP} \\ \frac{\Gamma \vdash A : \mathbf{U} \quad \Gamma, x:A \vdash B : \mathbf{U} \quad \Gamma \vdash f : \Pi(x:A)B \quad \Gamma \vdash a : A}{\Gamma \vdash f a : B[a]} \end{array}$$

Propositions

$$\begin{array}{c} \text{PROP} \\ \frac{\Gamma \text{ ctx}}{\Gamma \vdash \mathbf{Prop} : \mathbf{U}} \end{array} \qquad \begin{array}{c} \text{PROP-U} \\ \frac{\Gamma \vdash A : \mathbf{Prop}}{\Gamma \vdash A : \mathbf{U}} \end{array} \qquad \begin{array}{c} \text{PI-PROP} \\ \frac{\Gamma \vdash A : \mathbf{U} \quad \Gamma, x:A \vdash B : \mathbf{Prop}}{\Gamma \vdash \Pi(x:A)B : \mathbf{Prop}} \end{array}$$

4 Conversion rules

$$\begin{array}{c}
\text{REFL} \\
\frac{\Gamma \vdash M : A}{\Gamma \vdash M = M : A} \\
\\
\text{SYM} \\
\frac{\Gamma \vdash M = N : A}{\Gamma \vdash N = M : A} \\
\\
\text{TRANS} \\
\frac{\Gamma \vdash M = N : A \quad \Gamma \vdash N = P : A}{\Gamma \vdash M = P : A} \\
\\
\text{CONV-EQ} \\
\frac{\Gamma \vdash M = N : A \quad \Gamma \vdash A = B : \mathbb{U} \quad \Gamma \vdash B : \mathbb{U}}{\Gamma \vdash M = N : B} \\
\\
\text{PROP-IRR} \\
\frac{\Gamma \vdash A : \text{Prop} \quad \Gamma \vdash M : A \quad \Gamma \vdash N : A}{\Gamma \vdash M = N : A} \\
\\
\text{BETA} \\
\frac{\Gamma \vdash A : \mathbb{U} \quad \Gamma, x:A \vdash B : \mathbb{U} \quad \Gamma, x:A \vdash M : B \quad \Gamma \vdash a : A}{\Gamma \vdash (\lambda(x:A)M) a = M[a] : B[a]} \\
\\
\text{PI-CONG} \\
\frac{\Gamma \vdash A = A' : \mathbb{U} \quad \Gamma, x:A \vdash B = B' : \mathbb{U}}{\Gamma \vdash \Pi(x:A)B = \Pi(x:A')B' : \mathbb{U}}
\end{array}$$

Conversion rules for propositions

$$\begin{array}{c}
\text{PROP-U-EQ} \\
\frac{\Gamma \vdash M = N : \text{Prop}}{\Gamma \vdash M = N : \mathbb{U}} \\
\\
\text{PI-PROP-CONG} \\
\frac{\Gamma \vdash A = A' : \mathbb{U} \quad \Gamma, x:A \vdash B = B' : \text{Prop}}{\Gamma \vdash \Pi(x:A)B = \Pi(x:A')B' : \text{Prop}}
\end{array}$$

Note that in PI-CONG the codomain premise compares B and B' in the context extended by the *left* domain A , not by A' . This is the standard convention: since $A = A' : \mathbb{U}$ holds, one could equivalently use A' (via the conversion rule on contexts), but fixing the left side avoids a mutual dependency between the congruence rule and context conversion.

We write M^+ for the weakening of a term M by one fresh variable, i.e. M^+ is M viewed in the extended context $\Gamma, x:A$ where it does not depend on x .

$$\begin{array}{c}
\text{FUNEXT} \\
\frac{\Gamma \vdash A : \mathbb{U} \quad \Gamma, x:A \vdash f^+ x = g^+ x : B \quad \Gamma \vdash f : \Pi(x:A)B \quad \Gamma \vdash g : \Pi(x:A)B}{\Gamma \vdash f = g : \Pi(x:A)B} \\
\\
\text{APP-FUN} \\
\frac{\Gamma \vdash A : \mathbb{U} \quad \Gamma, x:A \vdash B : \mathbb{U} \quad \Gamma \vdash f = f' : \Pi(x:A)B \quad \Gamma \vdash a : A}{\Gamma \vdash f a = f' a : B[a]} \\
\\
\text{APP-ARG} \\
\frac{\Gamma \vdash A : \mathbb{U} \quad \Gamma, x:A \vdash B : \mathbb{U} \quad \Gamma \vdash f : \Pi(x:A)B \quad \Gamma \vdash a = a' : A}{\Gamma \vdash f a = f a' : B[a]}
\end{array}$$

Theorem 1 (Eta-conversion). *If $\Gamma \vdash A : \mathbb{U}$, $\Gamma, x:A \vdash B : \mathbb{U}$, and $\Gamma \vdash f : \Pi(x:A)B$, then*

$$\Gamma \vdash f = \lambda(x:A)f^+ x : \Pi(x:A)B.$$

Proof. By FUNEXT it suffices to show $\Gamma, x:A \vdash f^+ x = (\lambda(x:A)f^+ x)^+ x : B$. Since weakening distributes under λ , we have $(\lambda(x:A)M)^+ = \lambda(x:A^+)M^+$ (where the inner weakening acts on the next-to-last variable, skipping the binder). The right-hand side is therefore a β -redex. Applying BETA and the substitution identity $M^+[x_0] = M$ gives the result. Formalised in `EtaConversion.agda`. $\square$

5 Head reduction

We define head reduction $M \rightsquigarrow N$ as the compatible closure of β -reduction at the head:

$$\begin{array}{c} \text{HR-BETA} \\ \hline (\lambda(x:A)M) a \rightsquigarrow M[a] \end{array} \qquad \begin{array}{c} \text{HR-APP} \\ \hline \frac{M \rightsquigarrow M'}{M a \rightsquigarrow M' a} \end{array}$$

We write $M \rightsquigarrow^* N$ for the reflexive-transitive closure. The validity relation uses a wrapper $\text{Red } \Gamma \ M \ N \ A$, which carries only a proof of $M \rightsquigarrow^* N$; the context Γ and type A are phantom parameters kept for inference convenience. In [1], the reduction relation is defined as a subrelation of conversion, requiring both $M \rightsquigarrow^* N$ and $\Gamma \vdash M = N : A$. Here, reduction is purely syntactic: the conversion witnesses are instead stored directly at the leaves of the bundled validity relation (Val2/EqVal2), so the Red wrapper need not carry them.

6 Finite elements and the information ordering

Following Coquand–Huber [1], we interpret types and terms as ideal-domain elements, approximated from below by *finite elements*. The set FinEl and its auxiliary FinFun are defined by mutual induction:

$$\begin{aligned} u, v, a, b &::= \perp \mid \mathbf{U} \mid \mathbf{Prop} \mid \mathbf{Fun}(g) \mid \mathbf{Pi}(a, f) \\ g, f &\in \text{FinFun} = \text{List}(\text{FinEl} \times \text{FinEl}) \end{aligned}$$

Here $\perp$ is the least element (no information), $\mathbf{U}$ is the code for the universe, $\mathbf{Prop}$ is the code for the type of propositions, $\mathbf{Fun}(g)$ is a function element with graph g (a finite list of input–output pairs), and $\mathbf{Pi}(a, f)$ is a code for a Π -type with domain approximation a and codomain graph f . The codes $\mathbf{Prop}$ and $\mathbf{U}$ are incompatible in the ordering ($\mathbf{Prop} \sqcup \mathbf{U} = \perp$), but $\mathbf{Prop} : \mathbf{U}$ holds (propositions are types).

Binary supremum. A partial binary supremum $u \sqcup v$ is defined by case analysis: it combines compatible codes ($\mathbf{Pi}(a, f) \sqcup \mathbf{Pi}(b, g) = \mathbf{Pi}(a \sqcup b, f \mathbin{++} g)$, $\mathbf{Fun}(g) \sqcup \mathbf{Fun}(h) = \mathbf{Fun}(g \mathbin{++} h)$) and returns $\perp$ for incompatible pairs (e.g. $\mathbf{U} \sqcup \mathbf{Fun}(g) = \perp$).

Evaluation of finite functions. Given a finite function $f = [(u_1, v_1), \dots, (u_k, v_k)]$ and an input u , the sup-based evaluation collects all outputs whose keys are below u :

$$\text{ev}(f, u) = \bigsqcup \{v_i \mid u_i \leq u\}.$$

Information ordering. The ordering $u \leq v$ on FinEl is defined mutually with an ordering on finite functions. The key clauses are:

$$\begin{aligned} \perp &\leq v = \top \\ \mathbf{U} &\leq \mathbf{U} = \top \\ \mathbf{Prop} &\leq \mathbf{Prop} = \top \\ \mathbf{Fun}(g) &\leq \mathbf{Fun}(h) \iff g \leq_{\text{fin}} h \\ \mathbf{Pi}(a, f) &\leq \mathbf{Pi}(b, g) \iff a \leq b \wedge f \leq_{\text{fin}} g \end{aligned}$$

where the ordering on finite functions uses the *finite condition*:

$$g \leq_{\text{fin}} h \iff \forall (u_i, v_i) \in g. v_i \leq \text{ev}(h, u_i).$$

All other combinations (e.g. $\mathbf{U} \leq \mathbf{Fun}(g)$) are empty.

Coherence. A finite element is *coherent* ($\text{Coh}(u)$) when its internal data is consistent. Specifically, $\text{Coh}(\text{Pi}(a, f))$ requires $\text{Coh}(a)$ and that f is a *coherent finite function*: whenever two edges $(u_1, v_1), (u_2, v_2) \in f$ have compatible keys ($\text{Comp}(u_1, u_2)$), their values are also compatible ($\text{Comp}(v_1, v_2)$). This condition ensures that finite functions represent well-defined continuous maps between domains.

Finite membership. The *finite typing* relation $u : a$ (“ u is an element of the type coded by a ”) is defined by mutual recursion on u and a :

$$\begin{aligned}
\perp : a &\iff a : \mathbf{U} && (\text{every type has } \perp) \\
\mathbf{U} : \mathbf{U} &= \top \\
\text{Prop} : \mathbf{U} &= \top \\
\text{Fun}(g) : \text{Pi}(a, f) &\iff \text{FinMemFun}(g, a, f) \wedge \text{Coh}(\text{Fun}(g)) \wedge \text{Pi}(a, f) : \mathbf{U} \\
\text{Pi}(a, f) : \mathbf{U} &\iff a : \mathbf{U} \wedge \text{FinMemAllU}(f, a) \wedge \text{CohFunTail}(f) \\
\text{Pi}(a, f) : \text{Prop} &\iff a : \mathbf{U} \wedge \text{FinMemAllProp}(f, a) \wedge \text{CohFunTail}(f)
\end{aligned}$$

where $\text{FinMemAllProp}(f, a)$ checks that every edge $(a_i, b_i) \in f$ satisfies $a_i : a$ and $b_i : \text{Prop}$. where $\text{FinMemFun}(g, a, f)$ checks that every edge $(u_i, v_i) \in g$ satisfies $u_i : a$ and $v_i : \text{ev}(f, u_i)$, and $\text{FinMemAllU}(f, a)$ checks that every edge $(a_i, b_i) \in f$ satisfies $a_i : a$ and $b_i : \mathbf{U}$.

Key structural properties:

- $u : a$ implies $\text{Coh}(u)$, $\text{Coh}(a)$, and $a : \mathbf{U}$.
- Membership is monotone: if $u : a$ and $a \leq a'$ with $a' : \mathbf{U}$, then $u : a'$.
- *Prop collapse*: if $u : a$ and $a : \text{Prop}$, then $u = \perp$. (Inhabitants of propositions carry no information.)
- *Prop subtyping*: $u : \text{Prop}$ implies $u : \mathbf{U}$.
- $a \leq \text{Prop}$ implies $a = \perp$ or $a = \text{Prop}$.

7 Selections

A *selection* of a finite function f is a compatible subset of edges from f , yielding an aggregate key u and aggregate value v . Formally, $\text{Selection}(f, u, v)$ is an inductive type with three constructors:

$$\begin{array}{c}
\text{SEL-NIL} \\
\hline
\text{Selection}(\text{nil}, \perp, \perp)
\end{array}
\qquad
\begin{array}{c}
\text{SEL-SKIP} \\
\text{Selection}(g, u, v) \\
\hline
\text{Selection}((a_i, b_i)::g, u, v)
\end{array}$$

$$\begin{array}{c}
\text{SEL-TAKE} \\
\text{Comp}(a_i, u) \quad \text{Comp}(b_i, v) \quad \text{Selection}(g, u, v) \\
\hline
\text{Selection}((a_i, b_i)::g, a_i \sqcup u, b_i \sqcup v)
\end{array}$$

The compatibility conditions in SEL-TAKE ensure that the accumulated u and v remain coherent. A key property is that $\text{Selection}(f, u, v)$ implies $v \leq \text{ev}(f, u)$.

Why selections rather than individual edges? The evaluation $\text{ev}(f, u)$ is defined as a supremum: it aggregates the outputs of *all* edges in f whose keys lie below u . Working with single edges would

not capture this aggregate character. A selection encodes a finite compatible subfamily of edges whose joins approximate $\text{ev}(f, u)$ from below, and the compatibility check guarantees that these joins are well-defined (i.e. coherent). This is the mechanism that lets the relational semantics and the validity relation handle Π -types and λ -terms in a purely finite way, without ever computing an infinite supremum.

8 Relational semantics on finite approximants

Definition 2 (Evaluation relation). *Given a raw term M in context of size n , a finite environment ρ (mapping each variable to a FinEl), and a finite element u , the relation $\llbracket M \rrbracket_\rho \ni u$ (“ M evaluates to at least u in ρ ”) is defined by recursion on M :*

- **Variable:** $\llbracket x_i \rrbracket_\rho \ni u$ iff $\text{Coh}(u)$ and $u \leq \rho(i)$.
- **Universe:** $\llbracket U \rrbracket_\rho \ni u$ iff $\text{Coh}(u)$ and $u \leq U$.
- **Prop:** $\llbracket \text{Prop} \rrbracket_\rho \ni u$ iff $\text{Coh}(u)$ and $u \leq \text{Prop}$.
- **Application:** $\llbracket M N \rrbracket_\rho \ni \perp$ always holds. For non- $\perp$ output u : there exists v such that $\llbracket N \rrbracket_\rho \ni v$ and $\llbracket M \rrbracket_\rho \ni \text{Fun}([(v, u)])$. (One-edge form: the function M maps input v to output u .)
- **Lambda:** $\llbracket \lambda(x:A)M \rrbracket_\rho \ni \perp$ always holds. $\llbracket \lambda(x:A)M \rrbracket_\rho \ni \text{Fun}(g)$ requires a domain code a with $\text{Coh}(\text{Fun}(g))$, $a : U$, $\llbracket A \rrbracket_\rho \ni a$, and for every selection of g with key u and value v , there exists $x \leq u$ with $x : a$ and $\llbracket M \rrbracket_{\rho[x]} \ni v$.
- **Pi:** $\llbracket \Pi(x:A)B \rrbracket_\rho \ni \perp$ always holds. $\llbracket \Pi(x:A)B \rrbracket_\rho \ni \text{Pi}(a, f)$ requires $\text{Coh}(\text{Pi}(a, f))$, $\llbracket A \rrbracket_\rho \ni a$, and a witness a' with $\llbracket A \rrbracket_\rho \ni a'$ such that for every selection of f with key u and value v , there exists $x \leq u$ with $x : a'$ and $\llbracket B \rrbracket_{\rho[x]} \ni v$.

The relation is purely finite and has no reference to ideal semantic functions. Key properties:

- *Coherence extraction:* $\llbracket M \rrbracket_\rho \ni u$ implies $\text{Coh}(u)$.
- *Compatibility:* if $\llbracket M \rrbracket_\rho \ni u$ and $\llbracket M \rrbracket_\rho \ni v$ with $\text{Coh}(\rho)$, then $\text{Comp}(u, v)$.
- *Monotonicity in ρ :* if $\rho \leq \rho'$ pointwise and $\llbracket M \rrbracket_\rho \ni u$, then $\llbracket M \rrbracket_{\rho'} \ni u$.
- *Downward closure:* $\llbracket M \rrbracket_\rho \ni u$ and $u' \leq u$ with $\text{Coh}(u')$ implies $\llbracket M \rrbracket_\rho \ni u'$.
- *Sup closure:* $\llbracket M \rrbracket_\rho \ni u$ and $\llbracket M \rrbracket_\rho \ni v$ implies $\llbracket M \rrbracket_\rho \ni u \sqcup v$ (when the environment is coherent).

9 Soundness of the relational semantics

Definition 3 (Fitting environment). *An environment ρ fits a context Γ (written $\text{Fits}(\Gamma, \rho)$) when, for each variable $x_i : A_i$ in Γ , there exists a code a' such that $\rho(i) : a'$ and $\llbracket A_i \rrbracket_{\rho[i]} \ni a'$.*

Definition 4 (Typed evaluation). *Typed(M, A, ρ, u) asserts that u can be enlarged to some $u' \geq u$ which is a member of some code a' that A evaluates to: there exist u' and a' such that $u \leq u'$, $\llbracket M \rrbracket_\rho \ni u'$, $u' : a'$, and $\llbracket A \rrbracket_\rho \ni a'$.*

Theorem 5 (Typing soundness — Theorem 1 of [1]). *If $\Gamma \vdash M : A$ and $\text{Fits}(\Gamma, \rho)$, then for every u with $\llbracket M \rrbracket_\rho \ni u$, we have $\text{Typed}(M, A, \rho, u)$.*

Theorem 6 (Conversion soundness). *If $\Gamma \vdash M = N : A$ and $\text{Fits}(\Gamma, \rho)$, then:*

1. *Typing: $\text{Typed}(M, A, \rho, u)$ for all $\llbracket M \rrbracket_\rho \ni u$, and likewise for N .*
2. *Bidirectional transfer: for all u , $\llbracket M \rrbracket_\rho \ni u \iff \llbracket N \rrbracket_\rho \ni u$.*

The proofs proceed by mutual induction on the typing and conversion derivations. The proof-irrelevance case (PROP-IRR) uses the Prop collapse lemma: if $A : \text{Prop}$ and $M : A$, then any finite approximation u of M satisfies $u = \perp$. Since $\text{EvalRel}(N, \rho, \perp) = \top$ for all N , the bidirectional transfer is immediate. The β and function extensionality cases use the evaluation-substitution lemma: if $\llbracket M \rrbracket_{\rho[v]} \ni u$ and $\llbracket N \rrbracket_\rho \ni v$, then $\llbracket M[N/x] \rrbracket_\rho \ni u$ (and conversely). Formalised in `TypingSemantics.agda` and `LemmaForTS.agda`, with 0 postulates.

10 Logical relation (validity)

The validity relation is defined by mutual structural recursion on the type code $a \in \text{FinEl}$. It connects the syntactic judgments to the finite semantics.

Definition 7 (Validity). *For a context Γ , terms M, A , a realizer u , and a type code a , we define mutually:*

Term validity $\text{Val}(\Gamma, M, A, u, a)$ by case split on a :

- At $a = \perp$: always $\top$ (trivially valid).
- At $a = \text{U}$: reduces to type validity $\text{ValTy}(\Gamma, M, u)$.
- At $a = \text{Prop}$: always $\top$ (proof-irrelevance: all inhabitants of a proposition are trivially valid).
- At $a = \text{Pi}(b, f)$ with $u = \text{Fun}(g)$: a Σ -type containing
 1. type validity of A at $\text{Pi}(b, f)$ (head-reduces to some $\Pi A_0 B_0$),
 2. $\text{Coh}(\text{Fun}(g))$ and $\text{FinMemFun}(g, b, f)$,
 3. for every selection of g with key u' and value v , and argument N with $\text{Val}(\Gamma, N, A_0, u', b)$:

$$\text{Val}(\Gamma, M \ N, B_0[N], v, \text{ev}(f, u')).$$

- At $a = \text{Pi}(b, f)$ with $u \neq \text{Fun}(g)$: always $\top$.

Type validity $\text{ValTy}(\Gamma, M, u)$ at $u = \text{Pi}(b, f)$: a Σ -type containing terms A, B such that $M \rightsquigarrow^* \Pi(x:A)B$ in Γ , $\text{CohFunTail}(f)$, $\text{FinMemAllU}(f, b)$, type validity $\text{ValTy}(\Gamma, A, b)$, and for every selection of f with key u' and value v , and argument N with $\text{Val}(\Gamma, N, A, u', b)$: $\text{ValTy}(\Gamma, B[N], v)$.

Equality validity $\text{EqVal}(\Gamma, M, N, A, u, a)$ bundles validity of both M and N , plus type-equality data at Π -codes.

The key feature of this logical relation is that it recurses on the type code a rather than on syntactic types. Since $\text{ev}(f, u')$ is structurally smaller than $\text{Pi}(b, f)$ (its rank is bounded by $\text{rk}_{\text{fun}}(f) < \text{rk}(\text{Pi}(b, f))$), the recursion is well-founded despite the type-in-type universe.

Monotonicity properties. Three mutual monotonicity families are proved:

- *Downward in a:* if $a_0 \leq a_1$, $u : a_0$, $\text{Coh}(a_0)$, and $a_1 : \mathbf{U}$, then $\text{Val}(\Gamma, M, A, u, a_1) \Rightarrow \text{Val}(\Gamma, M, A, u, a_0)$.
- *Upward in a:* if $a_0 \leq a_1$, $u : a_0$, $u : a_1$, $\text{Coh}(a_0)$, $\text{Coh}(a_1)$, and $\text{ValTy}(\Gamma, A, a_1)$, then $\text{Val}(\Gamma, M, A, u, a_0) \Rightarrow \text{Val}(\Gamma, M, A, u, a_1)$.
- *Restriction in u:* if $u' \leq u$, $u' : a$, and $u : a$, then $\text{Val}(\Gamma, M, A, u, a) \Rightarrow \text{Val}(\Gamma, M, A, u', a)$.

Formalised in `Validity.agda` with 0 postulates.

11 Adequacy

The adequacy theorem is the bridge between the syntactic typing judgments and the semantic validity relation. It is proved using the two-substitution technique of [1], Theorem 2.

Definition 8 (Bundled validity). $\text{Val2}(\Gamma, M, A, u, a)$ refines Val by requiring $\Gamma \vdash M : A$ at the leaves (at $\perp$ -realizer or $\perp$ -code positions). Similarly, EqVal2 requires $\Gamma \vdash M = N : A$. These bundled versions are defined in `Validity2.agda`.

Theorem 9 (Adequacy — two-substitution form). *The following are proved by mutual induction on typing/conversion derivations, instantiated at two substitutions σ_1, σ_2 simultaneously:*

1. If $\Gamma \vdash M : A$, then for all codes u, a with $\llbracket M \rrbracket_\rho \ni u$ and $\llbracket A \rrbracket_\rho \ni a$:

$$\text{Val2}(\Delta, M\sigma, A\sigma, u, a).$$

2. If $\Gamma \vdash M = N : A$, then under the same conditions:

$$\text{EqVal2}(\Delta, M\sigma, N\sigma, A\sigma, u, a).$$

The proof uses the substitution lemma (well-typed substitutions preserve typing and conversion), context conversion, the soundness theorems 5 and 6, and the monotonicity properties of the validity relation. Key cases:

- **BETA:** uses the evaluation–substitution correspondence and β -expansion/contraction lemmas for Val2 .
- **FUNEXT:** uses weakening for EvalRel and the selection-based Pi clauses.
- **CONV:** uses the bidirectional transfer from conversion soundness and the downward/upward monotonicity of Val2 .
- **PROP-IRR:** by the Prop collapse lemma (Theorem 5 applied twice), any realizer u of a term in a Prop-type satisfies $u = \perp$, so EqVal2 at $\perp$ is trivially $\top$.
- **PI:** constructs a $\text{Pi}(b, f)$ -realizer by building selections from the domain and codomain adequacy hypotheses.

Formalised in `Adequacy2.agda` with 0 postulates.

12 Main result: Pi injectivity

Theorem 10 (Corollary 6; Coquand–Huber 2018, p. 661). *If $\Gamma \vdash A_0 = \Pi(x:B_1)F_1 : \mathbb{U}$, then there exist B_0, F_0 such that*

1. $A_0 \rightsquigarrow^* \Pi(x:B_0)F_0$,
2. $\Gamma \vdash B_0 = B_1 : \mathbb{U}$,
3. $\Gamma, x:B_0 \vdash F_0 = F_1 : \mathbb{U}$.

Corollary 11 (Pi–Pi injectivity). *If $\Gamma \vdash \Pi(x:A_0)B_0 = \Pi(x:A_1)B_1 : \mathbb{U}$, then*

$$\Gamma \vdash A_0 = A_1 : \mathbb{U} \quad \text{and} \quad \Gamma, x:A_0 \vdash B_0 = B_1 : \mathbb{U}.$$

Proof outline (Theorem). The proof proceeds by domain-theoretic adequacy.

1. From $\Gamma \vdash A_0 = \Pi B_1 F_1 : \mathbb{U}$, extract $\Gamma \vdash A_0 : \mathbb{U}$ by presupposition (SUBSTITUTIONLEMMA).
2. Apply the soundness theorem (CONVSOUND') at the *bottom environment* $\rho_\perp$ (mapping every variable to $\perp$) to obtain a bidirectional evaluation transfer: for every finite approximation u ,

$$\text{EvalRel}(\Pi B_1 F_1, \rho_\perp, u) \implies \text{EvalRel}(A_0, \rho_\perp, u).$$

3. Instantiate with $u = \text{PiCode}(\perp, \text{nil})$, which is coherent because $\text{CoherentFunTail}(\text{nil}) = \top$. The evaluation $\text{EvalRel}(\Pi B_1 F_1, \rho_\perp, \text{PiCode}(\perp, \text{nil}))$ is constructible trivially (domain $\perp$, empty function graph, vacuous selection body). Transfer gives $\text{EvalRel}(A_0, \rho_\perp, \text{PiCode}(\perp, \text{nil}))$.
4. Apply the bundled equality adequacy theorem (ADEQUACYEQSUB2) with the identity substitution at $\rho_\perp$ to obtain

$$\text{EqVal2 } \Gamma \ A_0 \ (\Pi B_1 F_1) \ \mathbb{U} \ (\text{PiCode}(\perp, \text{nil})) \ \text{UCode}.$$

5. This unfolds to EqValTyPi2 , a Σ -type containing:

- $\text{Red } \Gamma \ A_0 \ (\Pi B_0 F_0) \ \mathbb{U}$ (giving part 1),
- $\Gamma \vdash B_0 = B_1' : \mathbb{U}$ and $\Gamma, x:B_0 \vdash F_0 = F_1' : \mathbb{U}$,
- $\text{Red } \Gamma \ (\Pi B_1 F_1) \ (\Pi B_1' F_1') \ \mathbb{U}$.

Since Π is a head normal form, Red-unique-Pi forces $B_1' = B_1$ and $F_1' = F_1$, giving parts 2 and 3. □

Proof of Corollary. Apply the theorem to $\Gamma \vdash \Pi A_0 B_0 = \Pi A_1 B_1 : \mathbb{U}$. This yields B_0', F_0' with $\Pi A_0 B_0 \rightsquigarrow^* \Pi B_0' F_0'$. Since Π is a head normal form, Red-unique-Pi gives $B_0' = A_0$ and $F_0' = B_0$, so the domain and codomain conversions become $\Gamma \vdash A_0 = A_1 : \mathbb{U}$ and $\Gamma, x:A_0 \vdash B_0 = B_1 : \mathbb{U}$. □

Key intermediate lemmas. The proof above relies on several results that deserve explicit mention:

- *Presupposition* (SubstitutionLemma.agda): from a conversion judgment $\Gamma \vdash M = N : A$ one can extract $\Gamma \vdash M : A$, $\Gamma \vdash N : A$, and $\Gamma \vdash A : \mathbb{U}$. This is proved by mutual induction on typing and conversion, using the substitution and renaming lemmas.

- *Conversion soundness* (`convSound'` in `TypingSemantics.agda`): if $\Gamma \vdash M = N : A$ and $\text{EvalRel}(N, \rho, u)$, then $\text{EvalRel}(M, \rho, u)$ (and vice versa).
- *Bundled adequacy* (`adequacyEqSub2` in `Adequacy2.agda`): the bridge from `EvalRel` to the validity relation `EqVal2`, which packages both a semantic value and the corresponding syntactic typing/conversion derivation.
- *Reduction uniqueness for Π* (`Red-unique-Pi`): since Π is a head normal form, $\Pi A B \rightsquigarrow^* \Pi A' B'$ implies $A = A'$ and $B = B'$. This is the step that turns the “up to reduction” output of adequacy into exact equalities on the domain and codomain.

The entire proof is formalised in Agda with `--without-K --exact-split`, no `--type-in-type`, and no postulates (in the files on which `PiInjectivity.agda` depends).

13 Subject reduction

As an application of Pi injectivity, we prove subject reduction for single-step head reduction.

Theorem 12 (Subject reduction). *If $\Gamma \vdash M : A$ and $M \rightsquigarrow_1 N$ (single-step head reduction), then $\Gamma \vdash N : A$.*

Proof. By induction on the typing derivation, case-splitting on the reduction step.

- `CONV` and `PROP-U`: peel off and recurse.
- `APP` with $f \rightsquigarrow_1 f'$ (congruence under application): the induction hypothesis gives $\Gamma \vdash f' : \Pi(x:A)B$, and we rebuild with `APP`.
- `APP` with β -reduction $((\lambda(x:A')M') a \rightsquigarrow_1 M'[a])$: the key case. Given $\Gamma \vdash \lambda(x:A')M' : \Pi(x:A_0)B_0$ and $\Gamma \vdash a : A_0$, we extract $\Gamma, x:A_0 \vdash M' : B_0$ by walking the derivation through `CONV` (using Pi injectivity to connect the λ -annotation A' with the Π -domain A_0 and context/type-converting the body) and through `PROP-U` (which is eliminated: `piConv` applied to $\Gamma \vdash U = \Pi(x:A_0)B_0 : U$ yields $U \rightsquigarrow^* \Pi B'_0 F'_0$, which is impossible since U is a head normal form distinct from Π). Substitution then gives $\Gamma \vdash M'[a] : B_0[a]$.

Formalised in `SubjectReduction.agda` with 0 postulates. □

References

- [1] T. Coquand and S. Huber, “An adequacy theorem for dependent type theory,” *Archive for Mathematical Logic*, vol. 57, pp. 649–676, 2018.